

Calculator.dart

The image shows a mobile application interface titled "Last Name Activity 5" in a blue header bar. A red "DEBUG" banner is visible in the top right corner. The main area has a light purple background and contains two white input fields labeled "Number 1" and "Number 2". Below these fields are four colored buttons: a green "+" button, a red "-" button, an orange "*" button, and a blue "÷" button. At the bottom is a white rounded rectangle labeled "Result:" in purple text.

```

10 class _CalculatorState extends State<Calculator> {
11
12   TextEditingController num1Controller = TextEditingController();
13   TextEditingController num2Controller = TextEditingController();
14   String result = "";
15
16   void calculate(String operator){
17     double num1 = double.tryParse(num1Controller.text) ?? 0;
18     double num2 = double.tryParse(num2Controller.text) ?? 0;
19     double res = 0;
20
21     switch (operator){
22
23       case '+':
24         res = num1 + num2;
25         break;
26       case '-':
27         res = num1 - num2;
28         break;
29
30       case '*':
31         res = num1 * num2;
32         break;
33
34       case '/':
35         if(num2 != 0){
36           res = num1 / num2;
37         } else{
38           result = "Cannot divide by zero";
39           setState(() {
40
41             });
42           return;
43         }
44         break;
45
46     }
47     setState(() {
48       result = res.toStringAsFixed(2);
49     });
50   }
51

```

- num1Controller and num2Controller are controllers for the two text fields. They let you read the user's input.
- Reads the text from both text fields and converts them to double.
- Uses a switch to decide which operation to perform based on the button pressed.
- setState() tells Flutter to redraw the screen with the updated result.

```
51
52 Widget buildOperatorButton (String operator, Color color){
53     return ElevatedButton(style: ElevatedButton.styleFrom(
54         padding: EdgeInsets.symmetric(vertical: 18, horizontal: 24),
55         backgroundColor: color,
56         shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(12)),
57     ),
58     onPressed: () => calculate(operator),
59     child: Text(operator, style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold, color: Colors.white)),
60 ); // ElevatedButton
61 }
```

- A reusable widget that creates buttons for each operator (+, -, *, ÷).
- Each button calls calculate(operator) when pressed

```

63 Widget build(BuildContext context) {
64   return Scaffold(
65     appBar: AppBar(
66       title: Text("Last Name Activity 5"),
67       backgroundColor: Colors.blue,
68     ), // AppBar
69     body: Container(
70       padding: EdgeInsets.all(20),
71       decoration: BoxDecoration(
72         gradient: LinearGradient(colors: [Colors.purple.shade100, Colors.blue.shade100],
73           begin: Alignment.topLeft,
74           end: Alignment.bottomRight ), // LinearGradient
75       ), // BoxDecoration
76       child: Column(crossAxisAlignment: CrossAxisAlignment.stretch,
77         children: [TextField(
78           controller: num1Controller,
79           keyboardType: TextInputType.number,
80           decoration: InputDecoration(
81             labelText: "Number 1",
82             filled: true,
83             fillColor: Colors.white,
84             border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
85           ), // InputDecoration
86         ), // TextField
87         SizedBox(height: 15),
88         TextField(
89           controller: num2Controller,
90           keyboardType: TextInputType.number,
91           decoration: InputDecoration(
92             labelText: "Number 2",
93             filled: true,
94             fillColor: Colors.white,
95             border: OutlineInputBorder(borderRadius: BorderRadius.circular(12)),
96           ), // InputDecoration
97         ), // TextField
98         SizedBox(height: 20),
99         Row(
100           mainAxisAlignment: MainAxisAlignment.spaceEvenly,
101           children: [
102             buildOperatorButton('+', Colors.green),
103             buildOperatorButton('-', Colors.red),
104             buildOperatorButton('*', Colors.orange),
105             buildOperatorButton('/', Colors.blue),
106           ],
107         ), // Row

```

- Container wraps the main UI.
- padding: EdgeInsets.all(20) adds space from the screen edges.
- LinearGradient creates a background color transition from purple to blue.
- Row arranges the buttons horizontally.
- mainAxisAlignment.spaceEvenly spaces them evenly across the row.
- buildOperatorButton() is a custom function that creates a styled button for each operator.

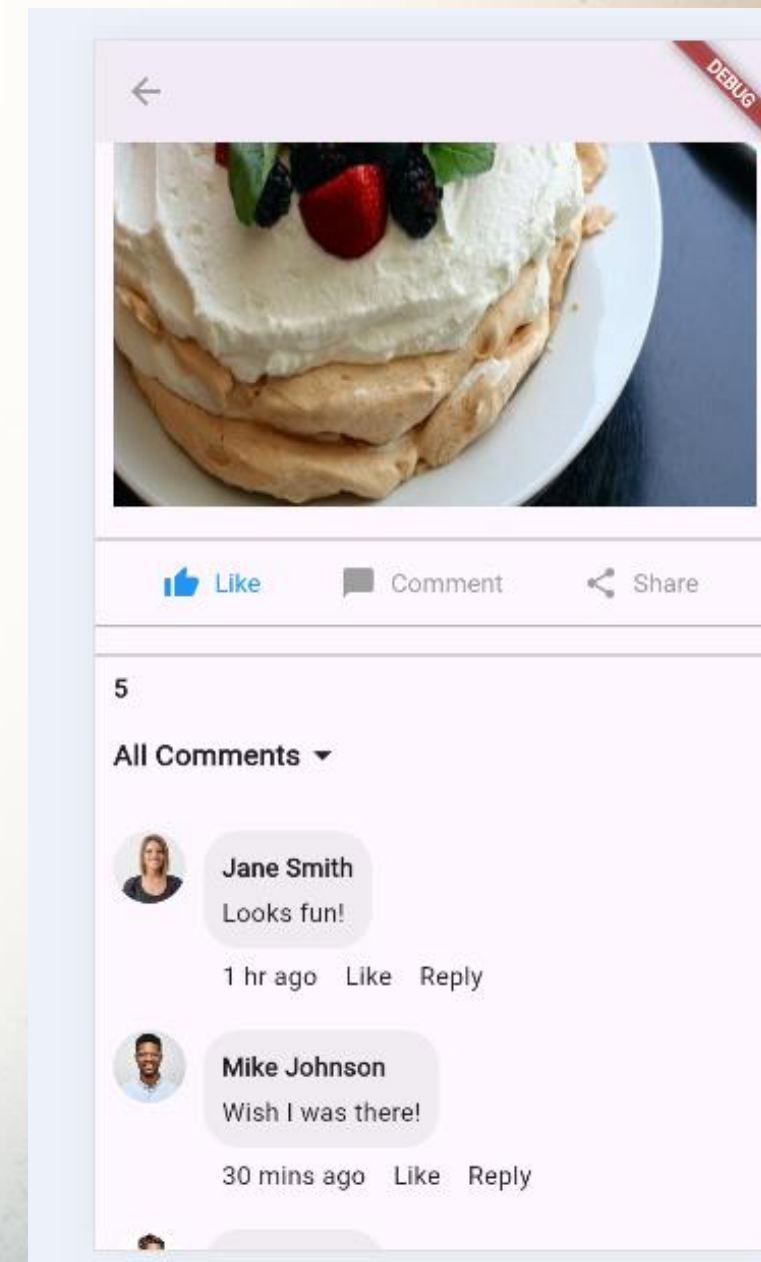
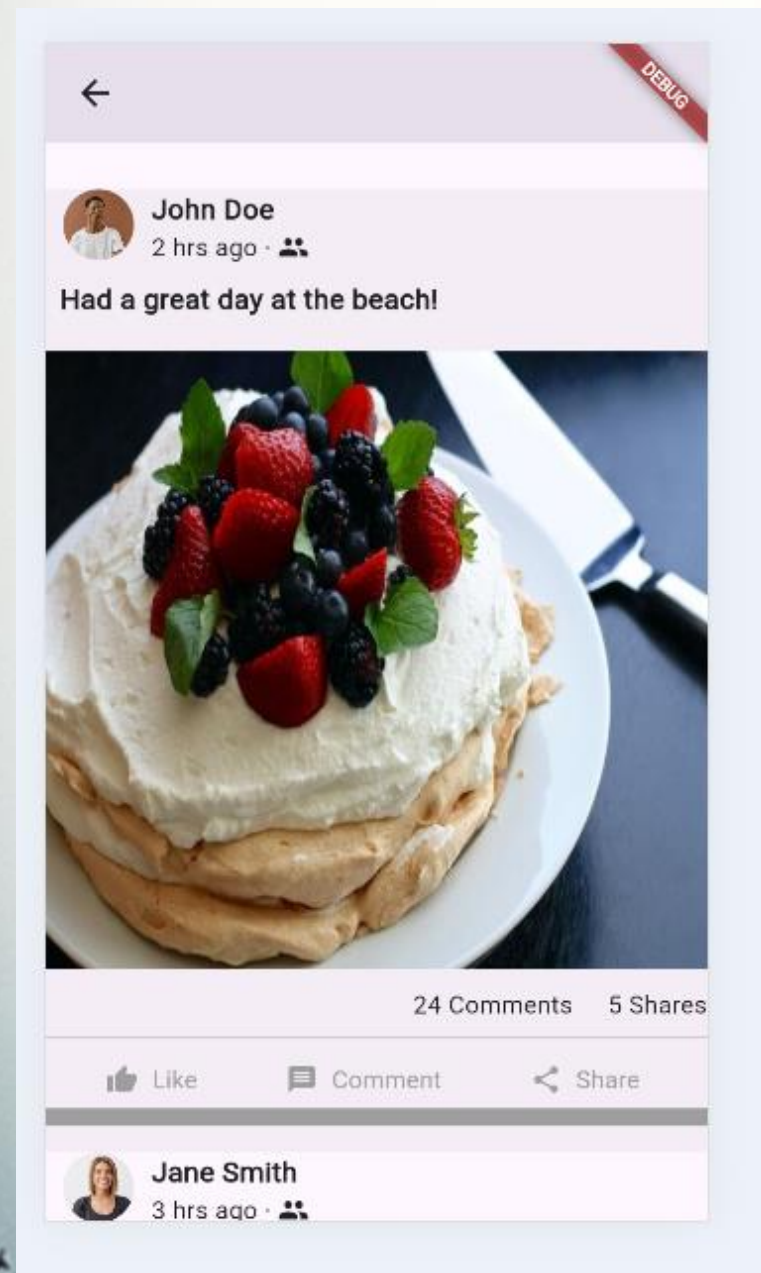
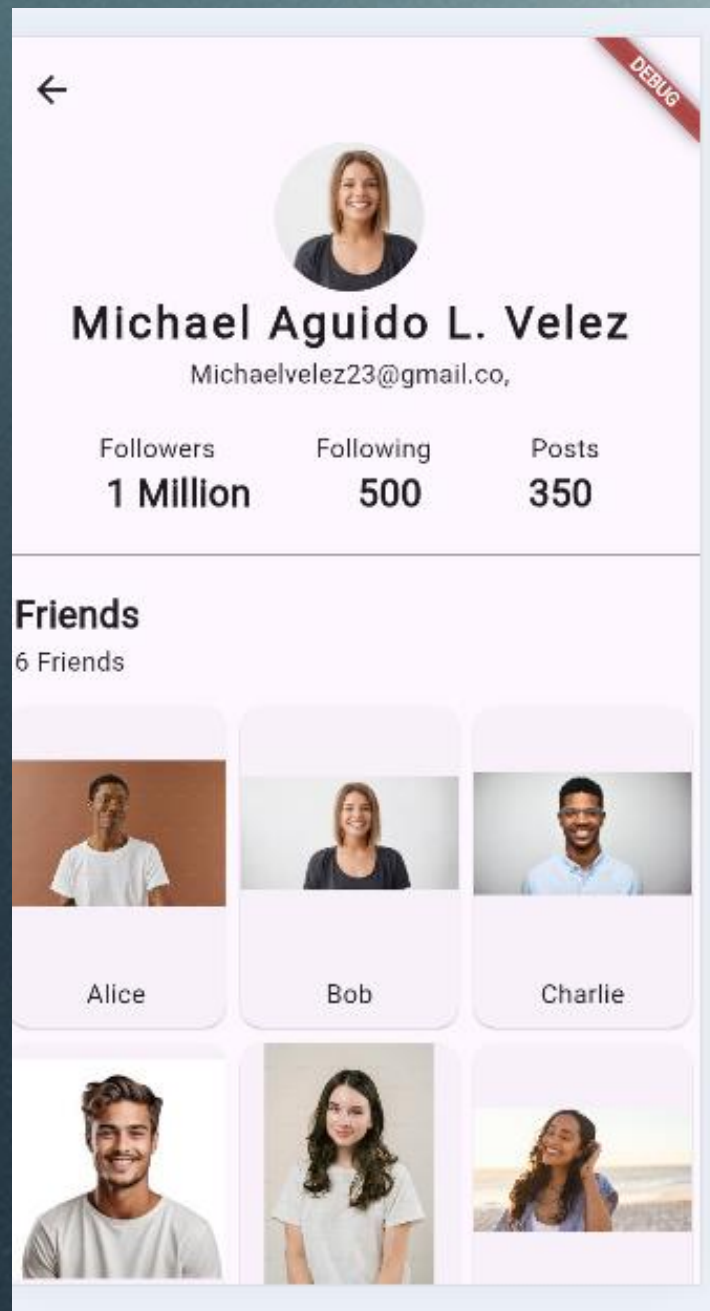
```

108     SizedBox(height: 25),
109     Container(
110       padding: EdgeInsets.all(20),
111       decoration: BoxDecoration(
112         color: Colors.white,
113         borderRadius: BorderRadius.circular(12),
114         boxShadow: [
115           BoxShadow(color: Colors.black, blurRadius: 8, offset: Offset(2, 4)),
116         ],
117       ), // BoxDecoration
118       child: Text("Result: $result" ,
119         style: TextStyle(fontSize: 28, fontWeight: FontWeight.bold, color: Colors.deepPurple),
120         textAlign: TextAlign.center,)), // Text
121     ), // Container
122   ],
123 ), // Column
124 ], // Container
125 ); // Scaffold
126 }
127 }

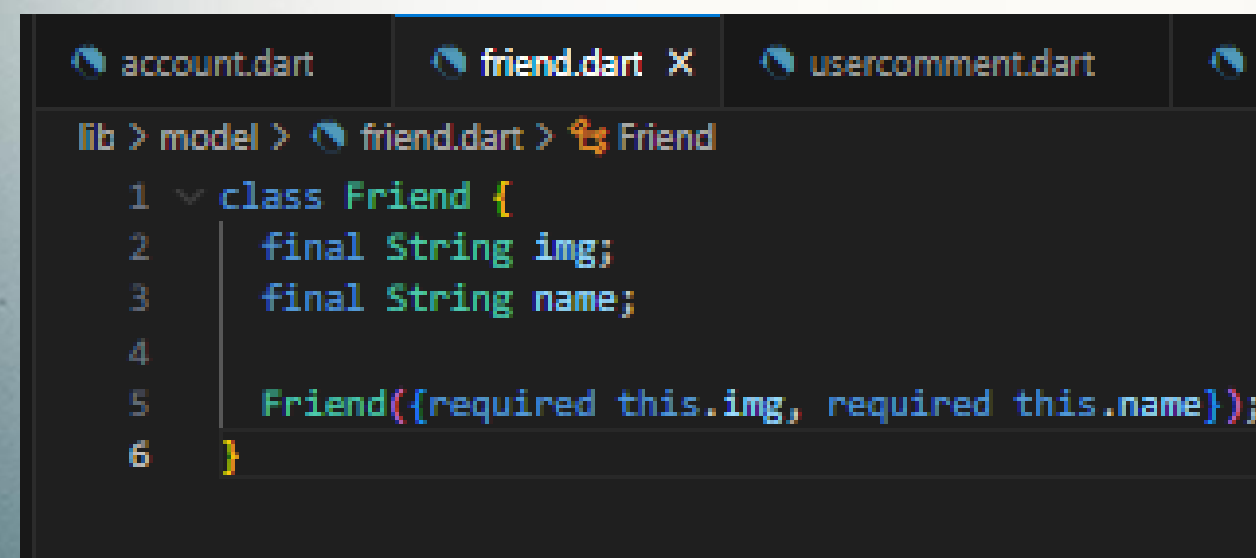
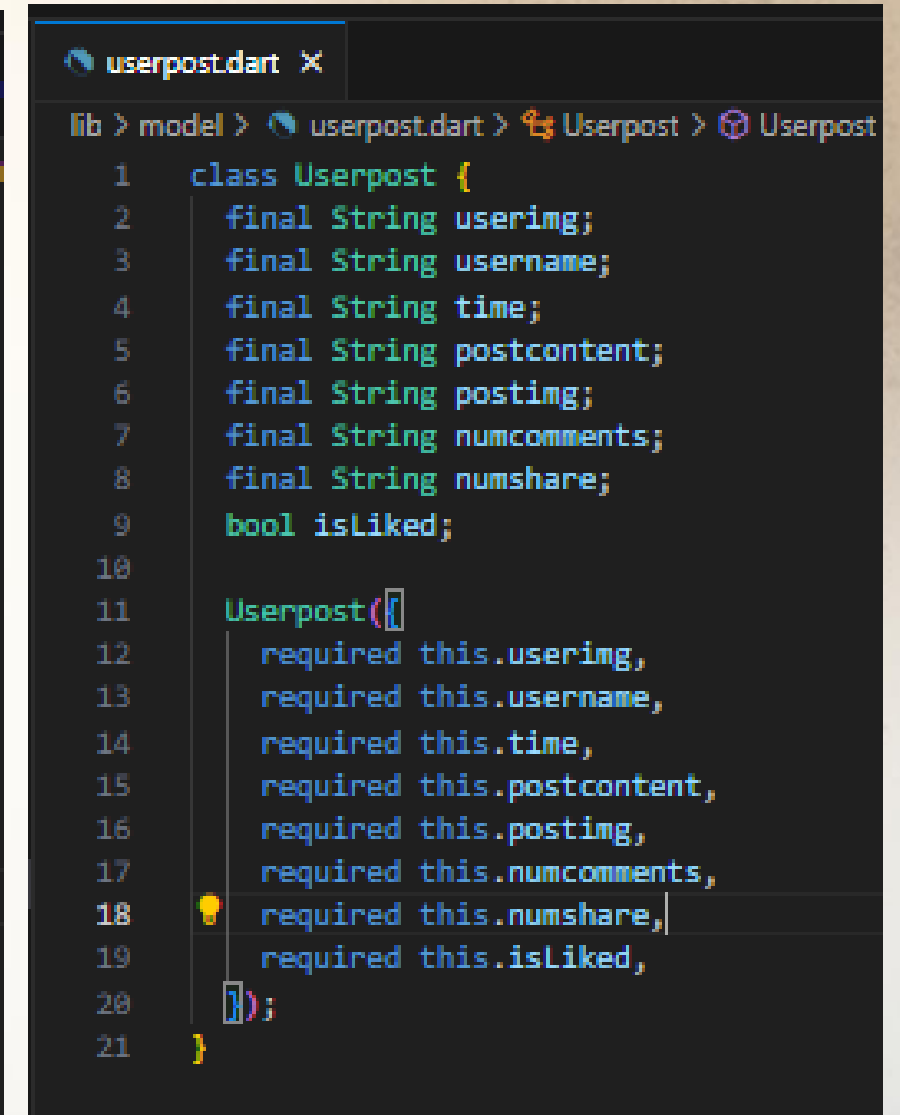
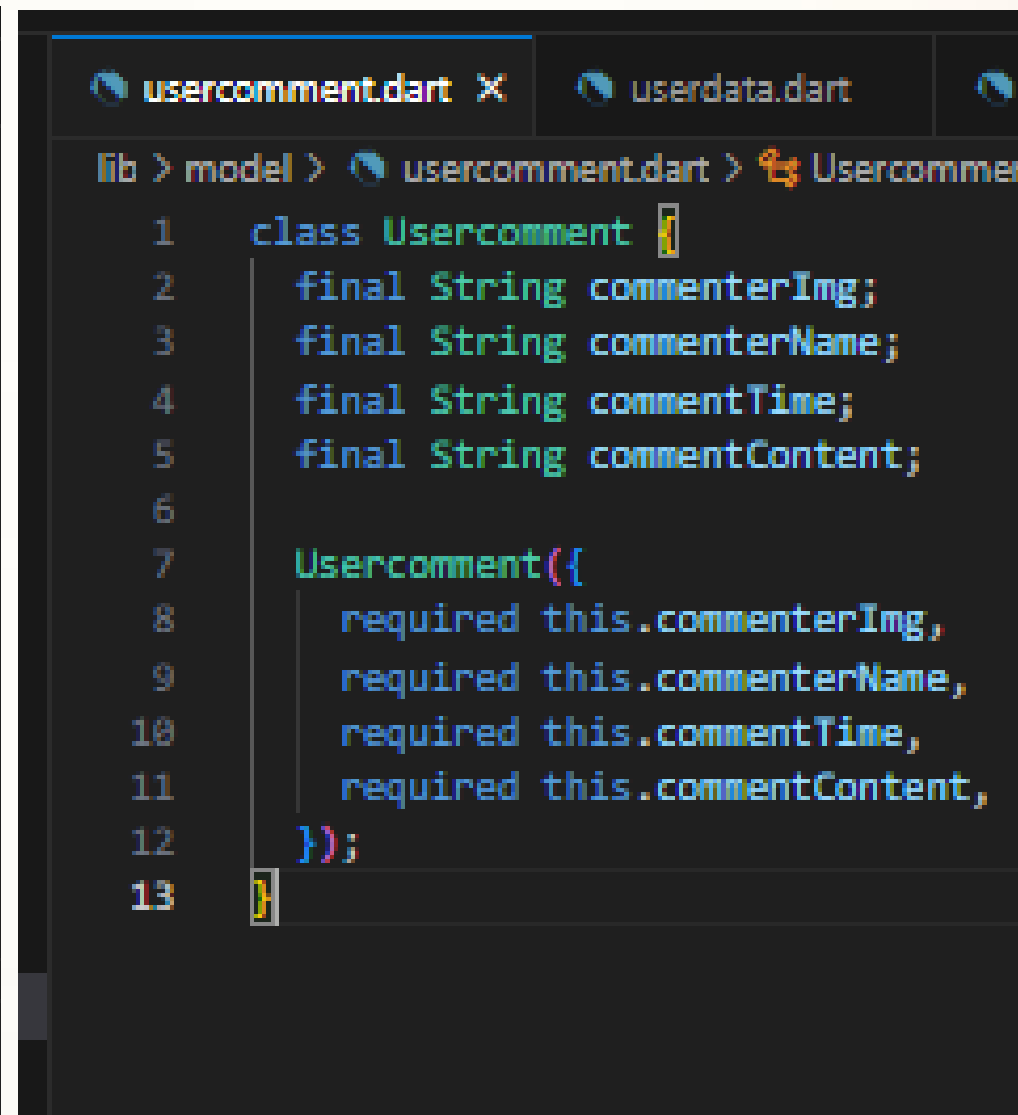
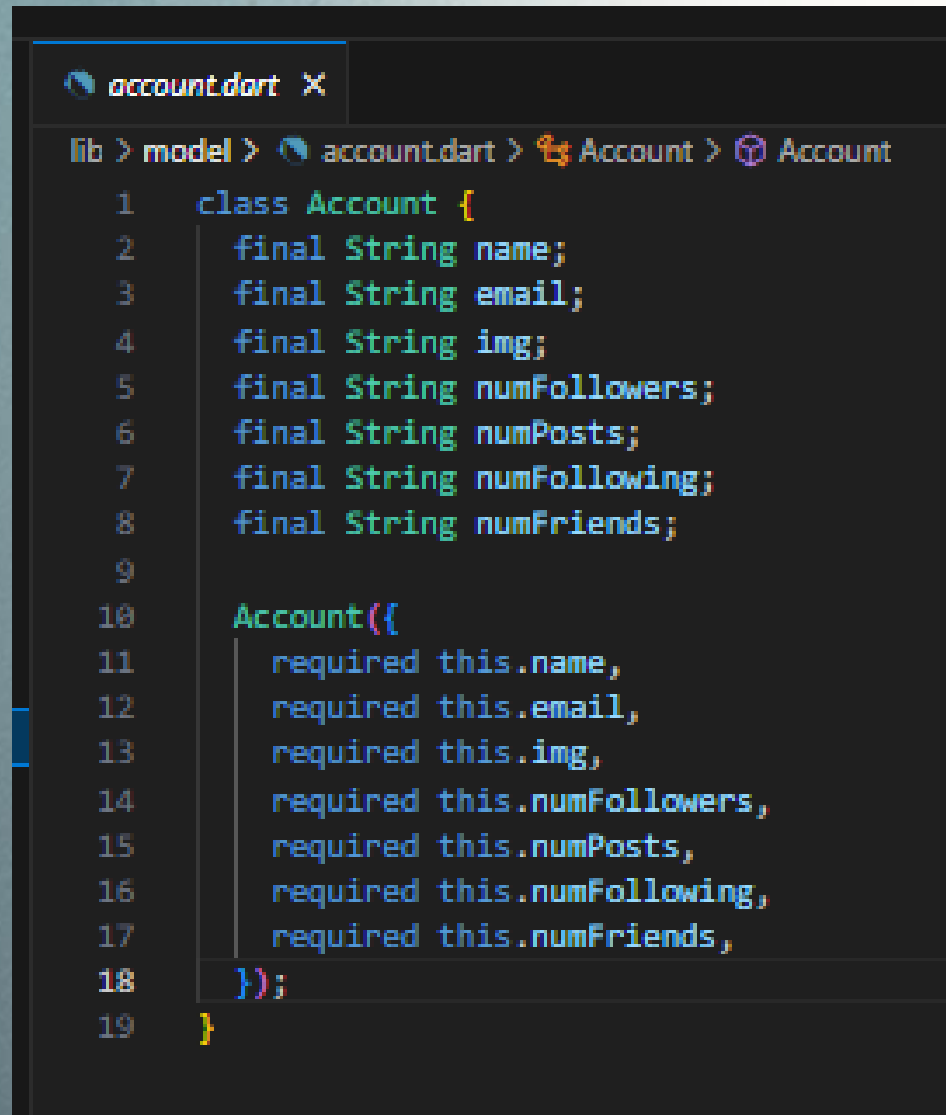
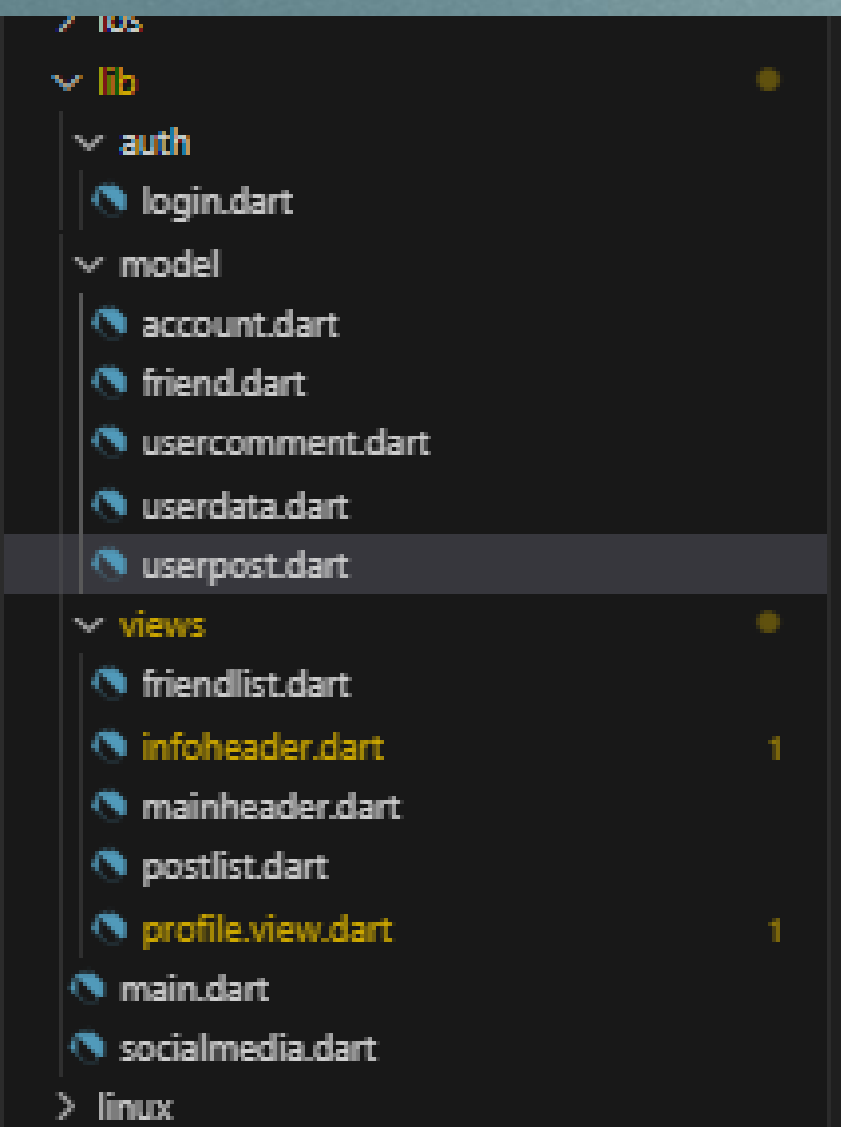
```

- Displays the result after calculation.
- We used boxShadow property to add shadow effect behind a widget

Social Media Clone



CREATE A MODEL FOLDER INSIDE THE LIB FOLDER.



- This file contains the data model for storing user information.
- A model class is used to structure and organize data in a clean and consistent way so it can be easily accessed and used throughout the app.

```

lib > model > userdata.dart > Userdata > userList
1  import 'friend.dart';
2  import 'usercomment.dart';
3  import 'userpost.dart';
4  import 'account.dart';
5
6  class Userdata {
7    List<Userpost> userList = [
8      Userpost(
9        usingimg: 'assets/person1.jfif',
10       username: 'John Doe',
11       time: '2 hrs ago',
12       postcontent: 'Had a great day at the beach!',
13       postimg: 'assets/pavlova.jpg',
14       numcomments: '24',
15       numshare: '5',
16       isLiked: false,
17     ),
18     Userpost(
19       usingimg: 'assets/person2.jfif',
20       username: 'Jane Smith',
21       time: '3 hrs ago',
22       postcontent: 'Loving the new cafe in town.',
23       postimg: 'assets/pavlova_main.jfif',
24       numcomments: '18',
25       numshare: '3',
26       isLiked: true,
27     ),
28     Userpost(
29       usingimg: 'assets/person3.jfif',
30       username: 'Mike Johnson',
31       time: '5 hrs ago',
32       postcontent: 'Just finished a 10k run!',
33       postimg: 'assets/pavlova.jpg',
34       numcomments: '30',
35       numshare: '7',
36       isLiked: false,
37     ),
38     Userpost(),
39     Userpost(
40       usingimg: 'assets/person4.jfif',
41       username: 'Johnson Stewart',
42       time: '5 hrs ago',
43       postcontent: 'Just finished a 10k run!',
44       postimg: 'assets/pavlova_main.jfif',
45       numcomments: '30',
46       numshare: '7',
47       isLiked: false,
48     ),
49   ];

```

```

51  List<Friend> friendList = [
52    Friend(img: 'assets/person1.jfif', name: 'Alice'),
53    Friend(img: 'assets/person2.jfif', name: 'Bob'),
54    Friend(img: 'assets/person3.jfif', name: 'Charlie'),
55    Friend(img: 'assets/person4.jfif', name: 'Diana'),
56    Friend(img: 'assets/person5.jfif', name: 'Ethan'),
57    Friend(img: 'assets/person6.jpg', name: 'Fiona'),
58  ];
59
60  List<Usercomment> commentList = [
61    Usercomment(
62      commenterImg: 'assets/person2.jfif',
63      commenterName: 'Jane Smith',
64      commentTime: '1 hr ago',
65      commentContent: 'Looks fun!',
66    ),
67    Usercomment(
68      commenterImg: 'assets/person3.jfif',
69      commenterName: 'Mike Johnson',
70      commentTime: '30 mins ago',
71      commentContent: 'Wish I was there!',
72    ),
73    Usercomment(
74      commenterImg: 'assets/person4.jfif',
75      commenterName: 'Emily Davis',
76      commentTime: '15 mins ago',
77      commentContent: 'Great photo!',
78    ),
79  ];
80
81  Account myUserAccount = Account(
82    name: 'Michael Aguido L. Velez',
83    email: 'Michaelvelez23@gmail.co,',
84    img: 'assets/person2.jfif',
85    numFollowers: '1 Million',
86    numPosts: '350',
87    numFollowing: '500',
88    numFriends: '400',
89  );
90 }

```

Create an Auth Folder Under Lib folder and create a dart file named login.dart

```
login.dart X
lib > auth > login.dart > _LoginFormState > dispose
1  import 'package:flutter/material.dart';
2  import 'package:flutter_application_7/socialmedia.dart';
3
4
5  class LoginForm extends StatefulWidget {
6    const LoginForm({super.key});
7
8    @override
9    State<LoginForm> createState() => _LoginFormState();
10 }
11
12 class _LoginFormState extends State<LoginForm> {
13   TextEditingController usernameController = TextEditingController();
14   TextEditingController passwordController = TextEditingController();
15   late String errorMessage;
16   late bool isError;
17
18   @override
19   void initState() {
20     errorMessage = "This is an errormessage";
21     isError = false;
22     super.initState();
23   }
24
25   @override
26   void dispose() {
27     super.dispose();
28   }
29
30   void checkLogin(username, password) {
31     setState(() {
32       if (username == "") {
33         errorMessage = "Please input your username!";
34         isError = true;
35       } else if (password == "") {
36         errorMessage = "Please input your password!";
37         isError = true;
38       } else {
39         errorMessage = "";
40         isError = false;
41       }
42     });
43   }
44 }
```

```
45 @override
46 Widget build(BuildContext context) {
47   return Scaffold(
48     body: Center(
49       child: Container(
50         margin: const EdgeInsets.symmetric(horizontal: 25),
51         child: Column(
52           mainAxisAlignment: MainAxisAlignment.center,
53           children: [
54             Row(
55               mainAxisAlignment: MainAxisAlignment.center,
56               children: [
57                 Image.asset(
58                   'assets/logo.png',
59                   height: 48,
60                   width: 48,
61                 ), // Image.asset
62                 const SizedBox(width: 12),
63                 Text('TIKITOKTOK', style: txtstyle),
64               ],
65             ), // Row
66             SizedBox(height: 15),
67             TextField(
68               controller: usernameController,
69               decoration: const InputDecoration(
70                 border: OutlineInputBorder(),
71                 labelText: 'Enter Username',
72                 prefixIcon: Icon(Icons.person),
73               ), // InputDecoration
74             ), // TextField
75             const SizedBox(height: 15),
76             TextField(
77               obscureText: true,
78               controller: passwordController,
79               decoration: const InputDecoration(
80                 border: OutlineInputBorder(),
81                 labelText: 'Enter Password',
82                 prefixIcon: Icon(Icons.password),
83               ), // InputDecoration
84             ), // TextField
85           ],
86         ),
87       ),
88     ),
89   );
90 }
```

```
85   SizedBox(height: 15),
86   ElevatedButton(
87     style: ElevatedButton.styleFrom(
88       minimumSize: const Size.fromHeight(50),
89       backgroundColor: Colors.blue,
90     ),
91     onPressed: () {
92       checkLogin(usernameController.text, passwordController.text);
93       if (!isError) {
94         Navigator.push(
95           context,
96           MaterialPageRoute(builder: (context) => Socialmedia()),
97         );
98       }
99     },
100     child: Text('LOGIN', style: txtstyle2),
101   ), // ElevatedButton
102 ],
103 ), // Column
104 ), // Container
105 ), // Center
106 ); // Scaffold
107 }
108
109
110
111 var txtstyle = const TextStyle(
112   fontWeight: FontWeight.bold,
113   letterSpacing: 2,
114   fontSize: 38,
115 );
116
117 var registertxtstyle = const TextStyle(
118   fontWeight: FontWeight.bold,
119   letterSpacing: 2,
120   fontSize: 14,
121 );
122
123 var errortxtstyle = const TextStyle(
124   fontWeight: FontWeight.bold,
125   letterSpacing: 1,
126   fontSize: 14,
127   color: Colors.red,
128 );
129
130 var txtstyle2 = const TextStyle(
131   fontWeight: FontWeight.bold,
132   letterSpacing: 2,
133   fontSize: 24,
134   color: Colors.white,
135 );
```

Create a View Folder Under Lib folder

```
friendlist.dart x
lib > views > friendlist.dart > Friendlist > build
1 import 'package:flutter/material.dart';
2 import 'package:flutter_application_7/model/friend.dart';
3
4 import '../model/userdata.dart';
5
6 // ignore: must_be_immutable
7 class Friendlist extends StatelessWidget {
8   Friendlist({super.key, required this.userdata});
9
10  final Userdata userdata;
11
12  var followTextStyle = const TextStyle(
13    fontSize: 20,
14    fontWeight: FontWeight.bold,
15  );
16
17  Widget friend(Friend friend) => Card(
18    child: Column(
19      children: [
20        Expanded(child: Image.asset(friend.img)),
21        Padding(padding: const EdgeInsets.all(8.0), child: Text(friend.name)),
22      ],
23    ), // Column
24  ); // Card
25
26  Widget friendsListGrid() => GridView.builder(
27    shrinkWrap: true,
28    physics: const BouncingScrollPhysics(),
29    gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
30      crossAxisCount: 3,
31      mainAxisExtent: 180,
32    ), // SliverGridDelegateWithFixedCrossAxisCount
33
34    itemCount: userdata.friendList.length,
35    itemBuilder: (BuildContext ctx, index) {
36      return friend(userdata.friendList[index]);
37    },
38  ); // GridView.builder
39
40  @override
41  Widget build(BuildContext context) {
42    return Column(
43      children: [
44        const SizedBox(height: 10),
45        Padding(
46          padding: const EdgeInsets.only(left: 8.0),
47          child: Row(children: [Text('Friends', style: followTextStyle)]),
48        ), // Padding
49        Padding(
50          padding: const EdgeInsets.only(left: 8.0),
51          child: Row(children: [Text('${userdata.friendList.length} Friends')]),
52        ), // Padding
53        const SizedBox(height: 10),
54        SizedBox(height: 380, child: friendsListGrid()),
55        SizedBox(height: 10, child: Container(color: Colors.grey)),
56      ],
57    ); // Column
58  }
59 }
```

```
infoheader.dart 1 x
lib > views > infoheader.dart > Infoheader
1 import 'package:flutter/material.dart';
2 import '../model/userdata.dart';
3
4 class Infoheader extends StatelessWidget {
5   Infoheader({super.key, required this.userdata});
6   final Userdata userdata;
7
8   var followTextStyle = const TextStyle(
9     fontSize: 20,
10    fontWeight: FontWeight.bold,
11  );
12
13  @override
14  Widget build(BuildContext context) {
15    return Column(
16      children: [
17        const Row(
18          mainAxisAlignment: MainAxisAlignment.spaceEvenly,
19          children: [Text('Followers'), Text('Following'), Text('Posts')],
20        ), // Row
21        Row(
22          mainAxisAlignment: MainAxisAlignment.spaceEvenly,
23          children: [
24            Text(userdata.myUserAccount.numFollowers, style: followTextStyle),
25            Text(userdata.myUserAccount.numFollowing, style: followTextStyle),
26            Text(userdata.myUserAccount.numPosts, style: followTextStyle),
27          ],
28        ), // Row
29        const SizedBox(height: 10),
30        const Divider(color: Colors.grey),
31      ],
32    ); // Column
33  }
34 }
```

Create a View Folder Under Lib folder

```
mainheader.dart X
lib > views > mainheader.dart > Mainheader > build
1 import 'package:flutter/material.dart';
2
3 import '../model/userdata.dart';
4
5 class Mainheader extends StatelessWidget {
6   const Mainheader({super.key, required this.userdata});
7
8   final Userdata userdata;
9
10  @override
11  Widget build(BuildContext context) {
12    return Column(
13      children: [
14        CircleAvatar(
15          backgroundImage: AssetImage(userdata.myUserAccount.img),
16          radius: 40,
17        ), // CircleAvatar
18        Text(
19          userdata.myUserAccount.name,
20          style: const TextStyle(
21            fontSize: 24,
22            fontWeight: FontWeight.bold,
23            letterSpacing: 2,
24          ), // TextStyle
25        ), // Text
26        Text(userdata.myUserAccount.email),
27        const SizedBox(height: 20),
28      ],
29    ); // Column
30  }
31 }
```

```
postlist.dart X
lib > views > postlist.dart > _PostlistState > build
15 class _PostlistState extends State<Postlist> {
103   Widget showPost(Userpost userPost) => Column(
112     const Divider(),
113     buttons(userPost),
114     SizedBox(height: 10, child: Container(color: Colors.grey)),
115     const SizedBox(height: 15),
116   ],
117 ); // Column
118
119 @override
120 Widget build(BuildContext context) {
121   return SingleChildScrollView(
122     scrollDirection: Axis.vertical,
123     child: ListView(
124       shrinkWrap: true,
125       children: widget.userdata.userList.map((userPost) {
126         return InkWell(
127           onTap: () {
128             gotoPage(context, ProfileView(userPost: userPost));
129           },
130           child: showPost(userPost),
131         ); // InkWell
132       }).toList(),
133     ), // ListView
134   ); // SingleChildScrollView
135 }
136 }
```

Continuation

```
105
106 Widget buttons(Userpost userPost) => Column(
107   mainAxisAlignment: MainAxisAlignment.start,
108   children: [
109     const Divider(color: Colors.grey),
110     Padding(
111       padding: const EdgeInsets.symmetric(horizontal: 10),
112       child: Row(
113         mainAxisAlignment: MainAxisAlignment.spaceEvenly,
114         children: [
115           TextButton.icon(
116             style: TextButton.styleFrom(
117               foregroundColor: userPost.isLiked ? Colors.grey : Colors.blue,
118             ),
119             onPressed: () {},
120             icon: const Icon(Icons.thumb_up, size: 20),
121             label: const Text('Like'),
122           ), // TextButton.icon
123           TextButton.icon(
124             style: TextButton.styleFrom(foregroundColor: Colors.grey),
125             onPressed: () {},
126             icon: const Icon(Icons.chat_bubble, size: 20),
127             label: const Text('Comment'),
128           ), // TextButton.icon
129           TextButton.icon(
130             style: TextButton.styleFrom(foregroundColor: Colors.grey),
131             onPressed: () {},
132             icon: const Icon(Icons.share, size: 20),
133             label: const Text('Share'),
134           ), // TextButton.icon
135         ],
136       ), // Row
137     ), // Padding
138     const Divider(color: Colors.grey),
139   ],
140 ); // Column
141
142
```

```
143
144 Widget userline(Userpost userPost) => Row(
145   crossAxisAlignment: CrossAxisAlignment.start,
146   mainAxisAlignment: MainAxisAlignment.start,
147   children: [
148     Container(
149       padding: const EdgeInsets.symmetric(horizontal: 10),
150       child: CircleAvatar(
151         backgroundImage: AssetImage(userPost.userimg),
152         radius: 20,
153       ), // CircleAvatar
154     ), // Container
155     Column(
156       mainAxisAlignment: MainAxisAlignment.start,
157       crossAxisAlignment: CrossAxisAlignment.start,
158       children: [
159         Text(userPost.username, style: nametxtStyle),
160         const SizedBox(height: 5),
161         Row(
162           children: [
163             Text(userPost.time),
164             const Text('.'),
165             const Icon(Icons.group, size: 15, color: Colors.grey),
166           ], // Row
167         ), // Column
168       ],
169     ), // Row
170 ); // Row
171
172 Widget postimage(Userpost userPost) => Padding(
173   padding: const EdgeInsets.all(10.0),
174   child: Column(
175     mainAxisAlignment: MainAxisAlignment.center,
176     children: [
177       Row(children: [Text(userPost.postcontent)]),
178       const SizedBox(height: 15),
179       Container(
180         height: 350,
181         decoration: BoxDecoration(
182           image: DecorationImage(
183             image: AssetImage(userPost.postimg),
184             fit: BoxFit.fill,
185           ), // DecorationImage
186         ), // BoxDecoration
187       ), // Container
188     ],
189   ), // Column
190 ); // Padding
191
```

Continuation

```
192 @override
193 Widget build(BuildContext context) {
194   return Scaffold(
195     appBar: AppBar(
196       backgroundColor: Colors.transparent,
197       bottomOpacity: 0.0,
198       elevation: 0.0,
199       leading: IconButton(
200         onPressed: () {
201           Navigator.pop(context);
202         },
203         icon: const Icon(Icons.arrow_back, color: Colors.grey),
204       ), // IconButton
205     ), // AppBar
206     body: ListView(
207       shrinkWrap: true,
208       children: [
209         userline(userPost),
210         postimage(userPost),
211         buttons(userPost),
212         commenters(userPost),
213         ListView(
214           shrinkWrap: true,
215           children: userData.commentList
216             .map((userComment) => userpostdetails(userComment))
217             .toList(),
218         ), // ListView
219       ],
220     ), // ListView
221   ); // Scaffold
222 }
223 }
```

Socialmedia.dart

```
lib > socialmedia.dart > _SocialmediaState > build
1 import 'package:flutter/material.dart';
2 import 'package:flutter_application_7/model/userdata.dart';
3 import 'package:flutter_application_7/views/friendlist.dart';
4 import 'package:flutter_application_7/views/infoheader.dart';
5 import 'package:flutter_application_7/views/mainheader.dart';
6 import 'package:flutter_application_7/views/postlist.dart';
7
8
9 class Socialmedia extends StatefulWidget {
10   const Socialmedia({super.key});
11
12   @override
13   State<Socialmedia> createState() => _SocialmediaState();
14 }
15
16 class _SocialmediaState extends State<Socialmedia> {
17   Userdata userdata = Userdata();
18
19   var followTextStyle = const TextStyle(
20     fontSize: 20,
21     fontWeight: FontWeight.bold,
22   );
23
24   @override
25   Widget build(BuildContext context) {
26     return Scaffold(
27       appBar: AppBar(
28         leading: IconButton(
29           onPressed: () {
30             Navigator.pop(context);
31           },
32           icon: Icon(Icons.arrow_back),
33         ), // IconButton
34       ), // AppBar
35       body: ListView(
36         shrinkWrap: true,
37         children: [
38           Mainheader(userdata: userdata),
39           Infoheader(userdata: userdata),
40           Friendlist(userdata: userdata),
41           const SizedBox(height: 20),
42           Padding(
43             padding: const EdgeInsets.only(left: 8.0),
44             child: Row(children: [Text('Posts', style: followTextStyle)]),
45           ), // Padding
46           const SizedBox(height: 20),
47           Postlist(userdata: userdata),
48         ],
49       ), // ListView
50     ); // Scaffold
51   }
52 }
```

Main.dart

```
lib > main.dart > MyApp > build
1 import 'package:flutter/material.dart';
2 import 'package:flutter_application_7/auth/login.dart';
3
4 Run | Debug | Profile
5 void main() {
6   runApp(const MyApp());
7 }
8
9 class MyApp extends StatelessWidget {
10   const MyApp({super.key});
11
12   @override
13   Widget build(BuildContext context) {
14     return MaterialApp(
15       title: 'Flutter Demo',
16       theme: ThemeData(
17         colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
18       ), // ThemeData
19       home: const LoginForm(),
20     ); // MaterialApp
21   }
22
23 }
```